



EcoFix AI

Empowering the Circular
Economy through
Agentic AI

AI for Sustainability Capstone Project (SDG 12)

The E-Waste Tsunami

53.6 million

metric tonnes of global e-waste generated annually.

The equivalent weight of
4,500 Eiffel Towers
discarded every single year.

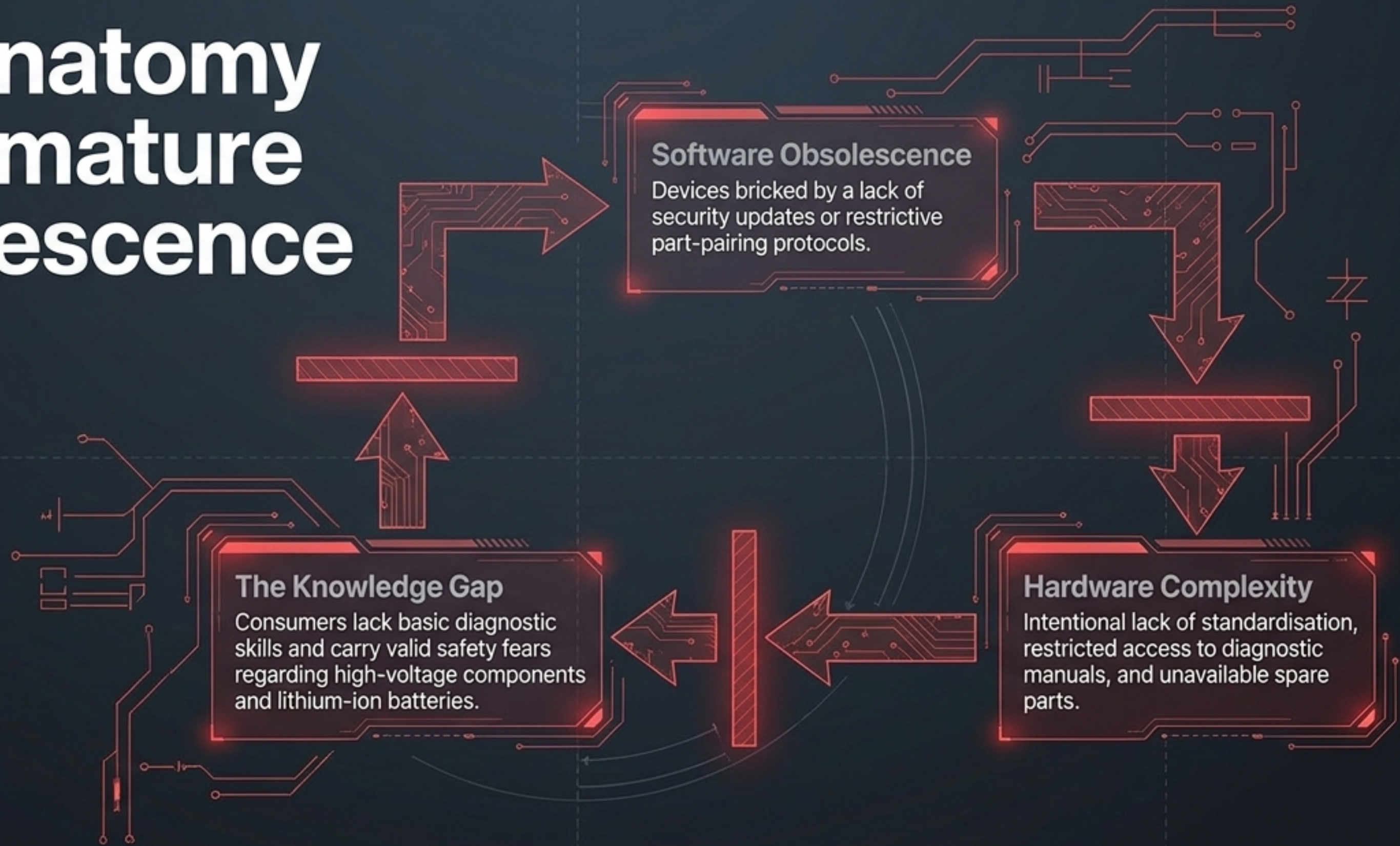
The UK Context:

21 kg of e-waste per capita.

It is currently the world's fastest-growing waste stream, yet the global recycling rate languishes at approximately **21%.**



The Anatomy of Premature Obsolescence



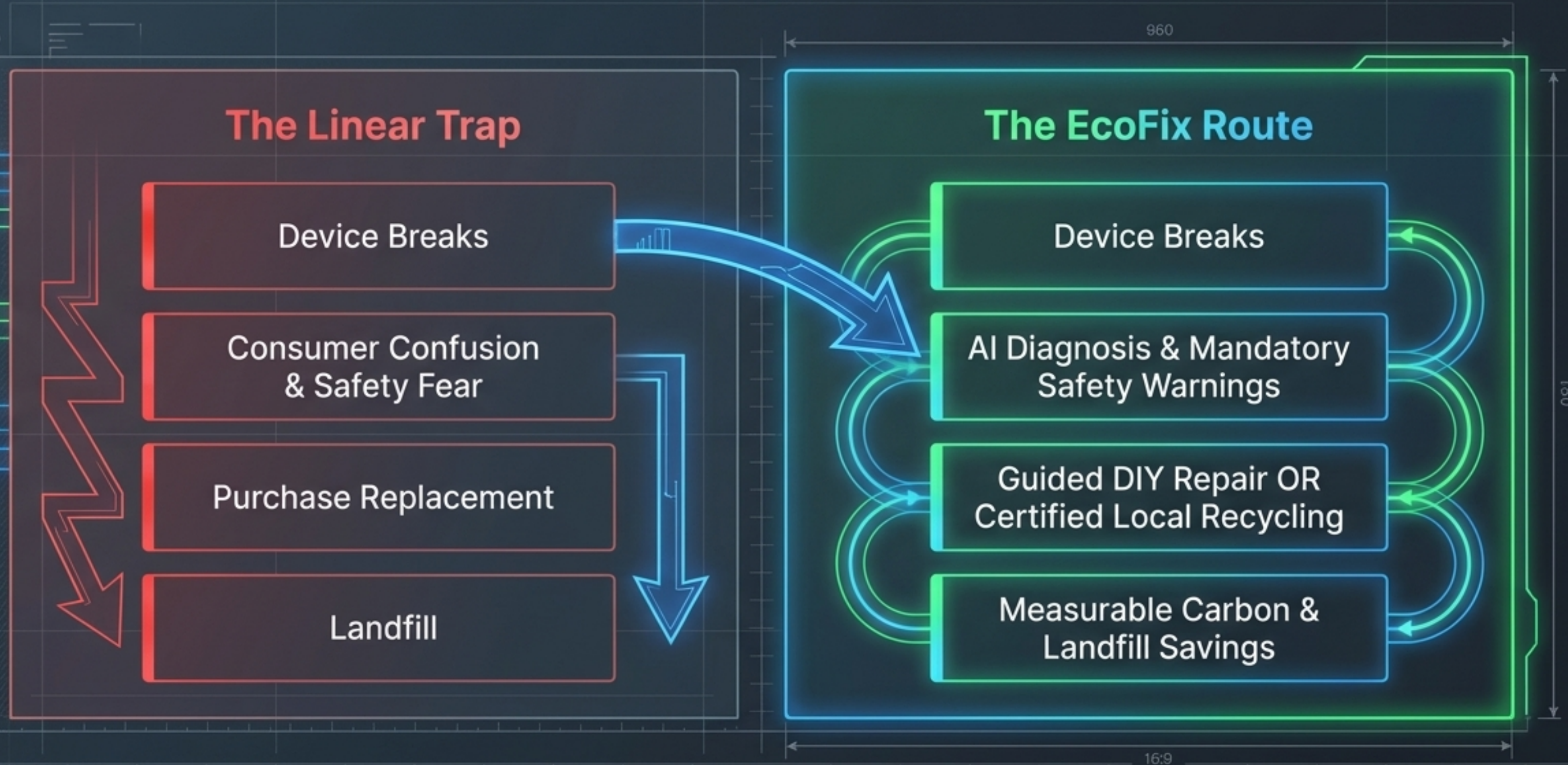
Synthesis Insight: The current economic system is fundamentally designed for disposal, not repair.

Activating the Circular Economy



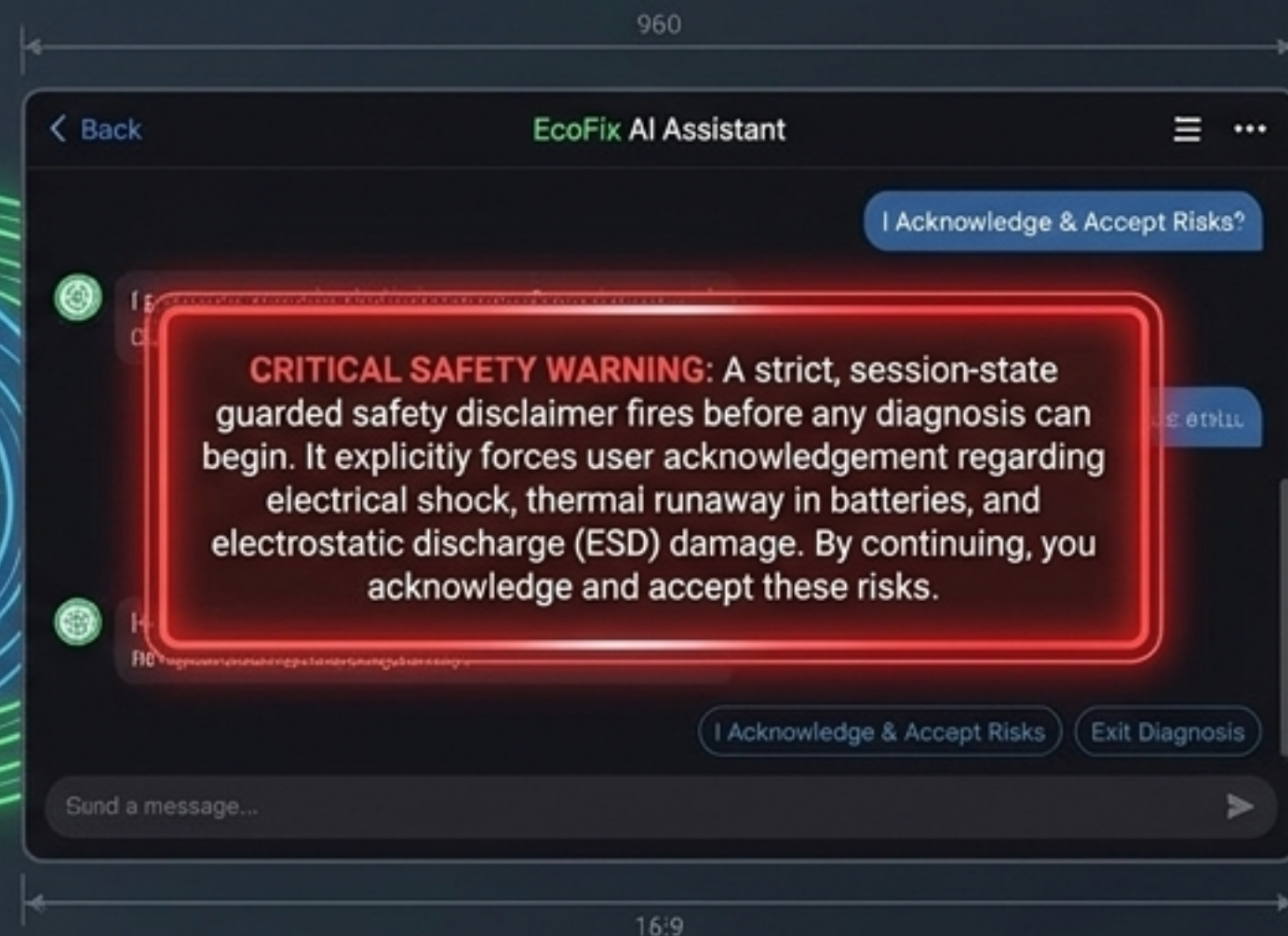
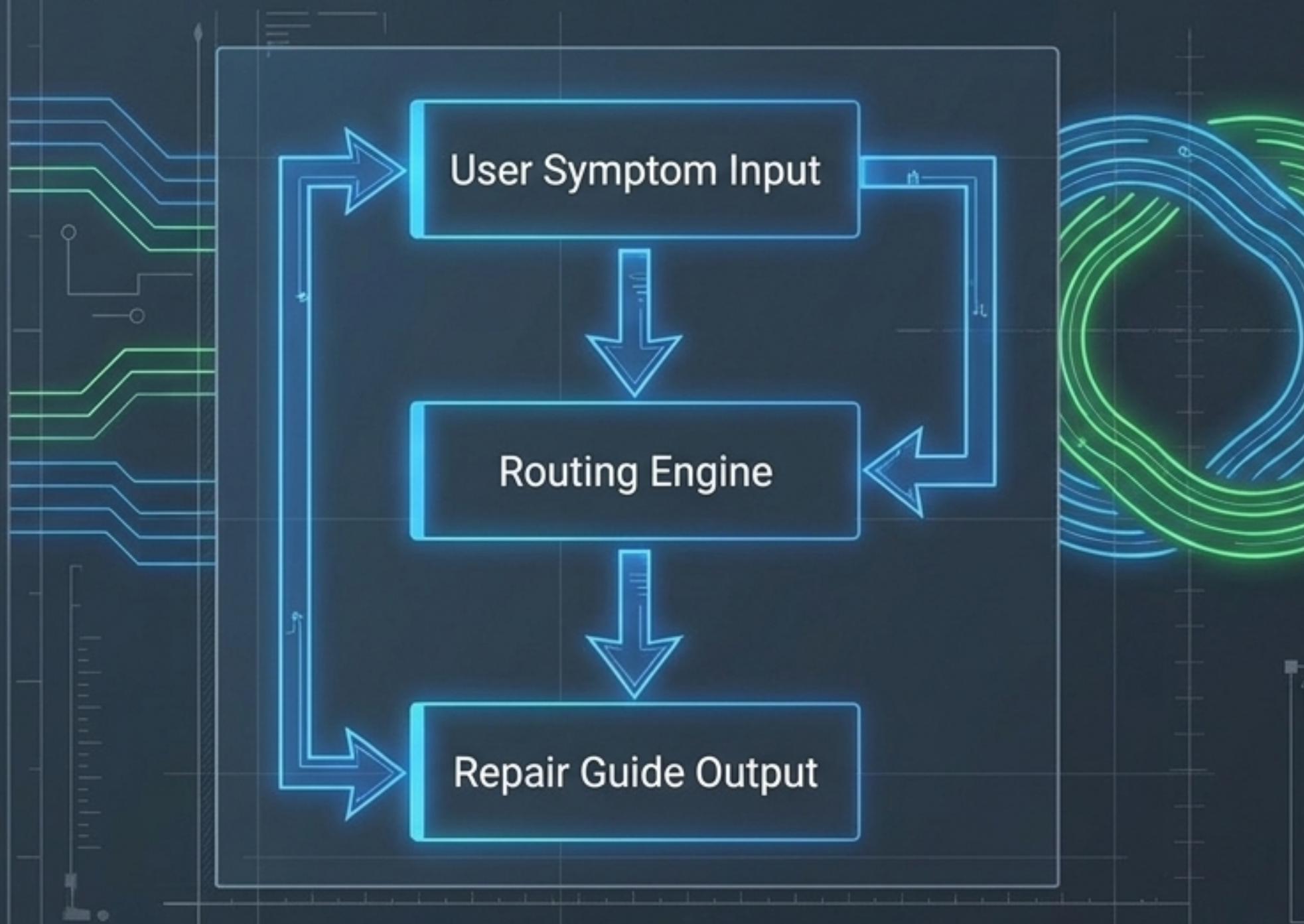
Re-routing the Consumer Journey

Target Users: DIY repairers, eco-conscious consumers, and everyday owners of household appliances.



The Diagnostic Assistant

Functionality: An intelligent chatbot that processes device symptoms and returns step-by-step DIY repair guides, difficulty ratings, and required tools.



Responsible AI Guardrail: A strict, session-state guarded safety disclaimer fires before any diagnosis can begin. It explicitly forces user acknowledgement regarding electrical shock, thermal runaway in batteries, and electrostatic discharge (ESD) damage.

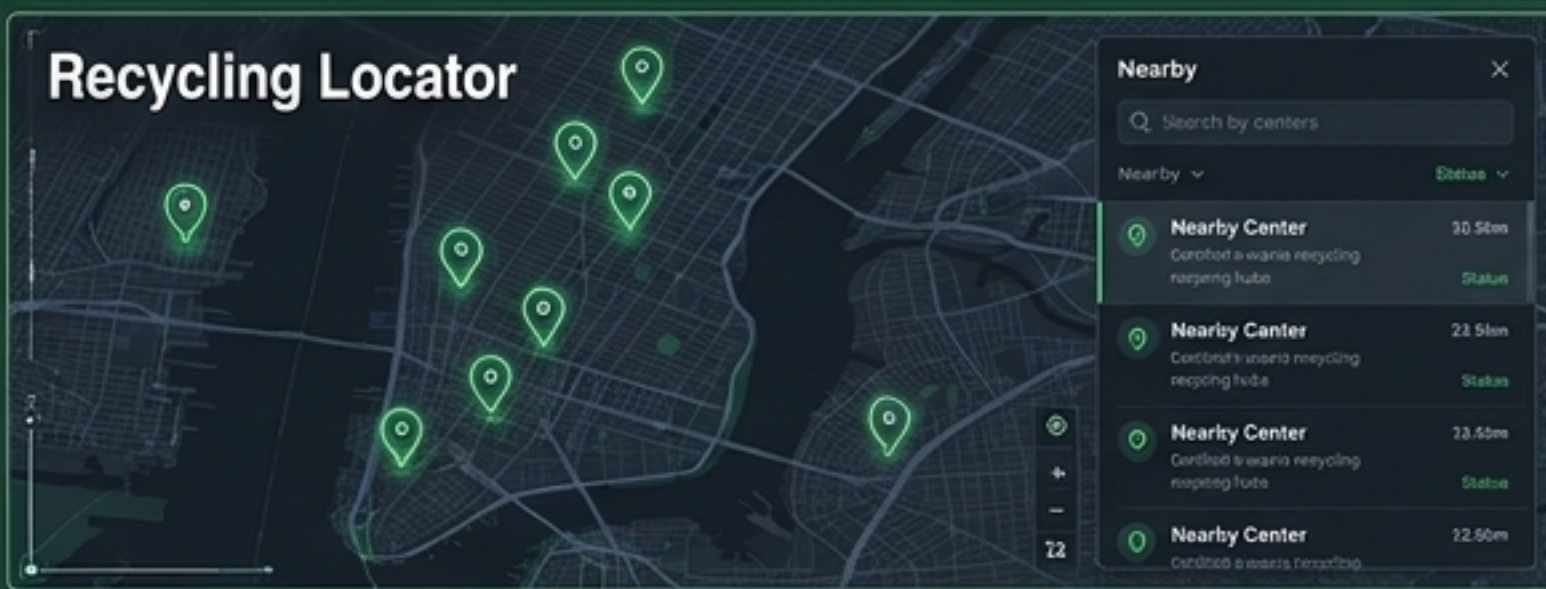
Hardware Education & End-of-Life Routing

Hardware & Tools Guide



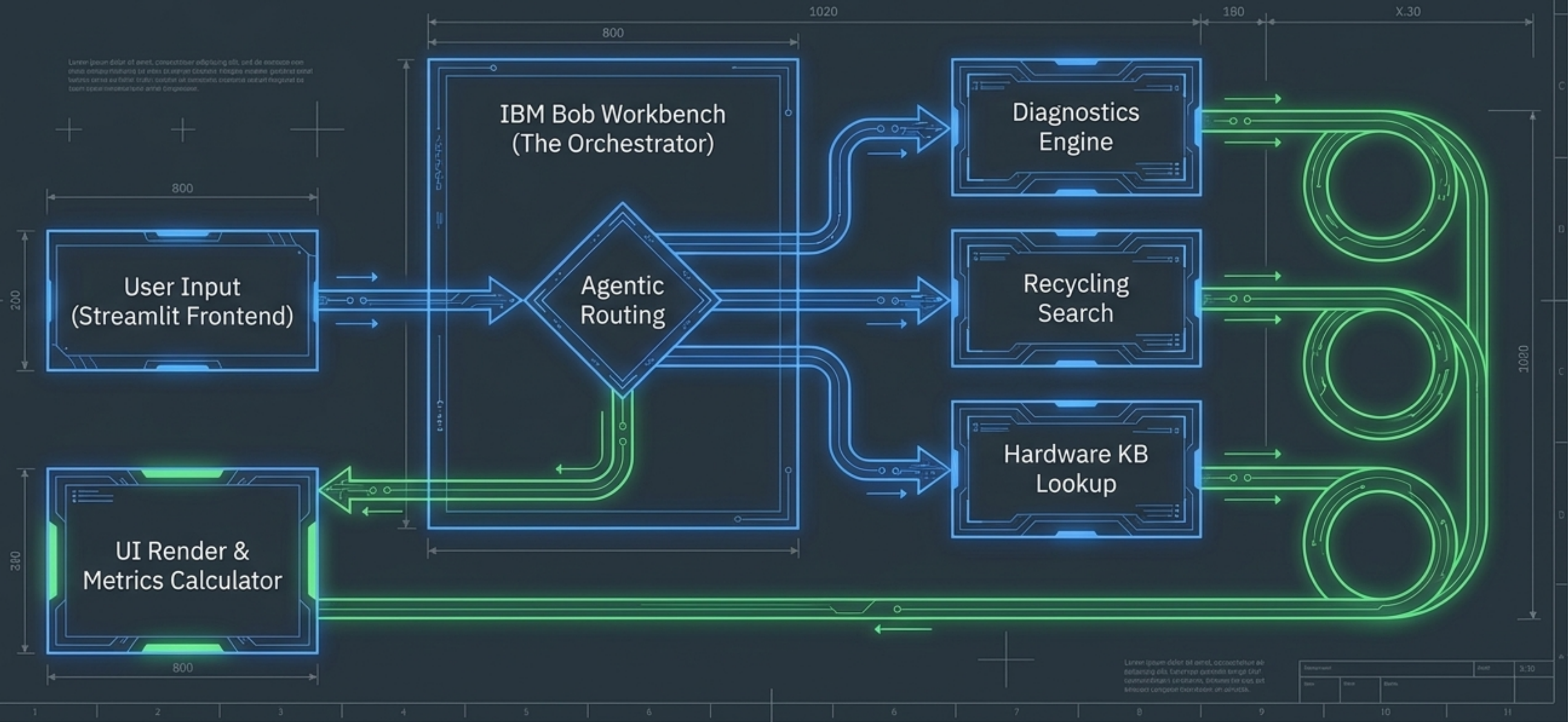
Hardware & Tools Guide: An interactive encyclopaedia breaking down repair tools (multimeter modes, soldering irons) and appliance circuit components (IPM ICs, relays). Includes a dynamic Ohm's Law and Power calculator for component testing.

Recycling Locator



Recycling Locator: When devices are beyond repair, the app routes users to certified e-waste hubs. Powered by a primary CSV data fallback for reliability, with an optional live integration via the Earth911 API.

Agentic AI Orchestration



The Technology Stack

Presentation Layer

Streamlit (Multi-page configuration with session-state management).

Orchestration Layer

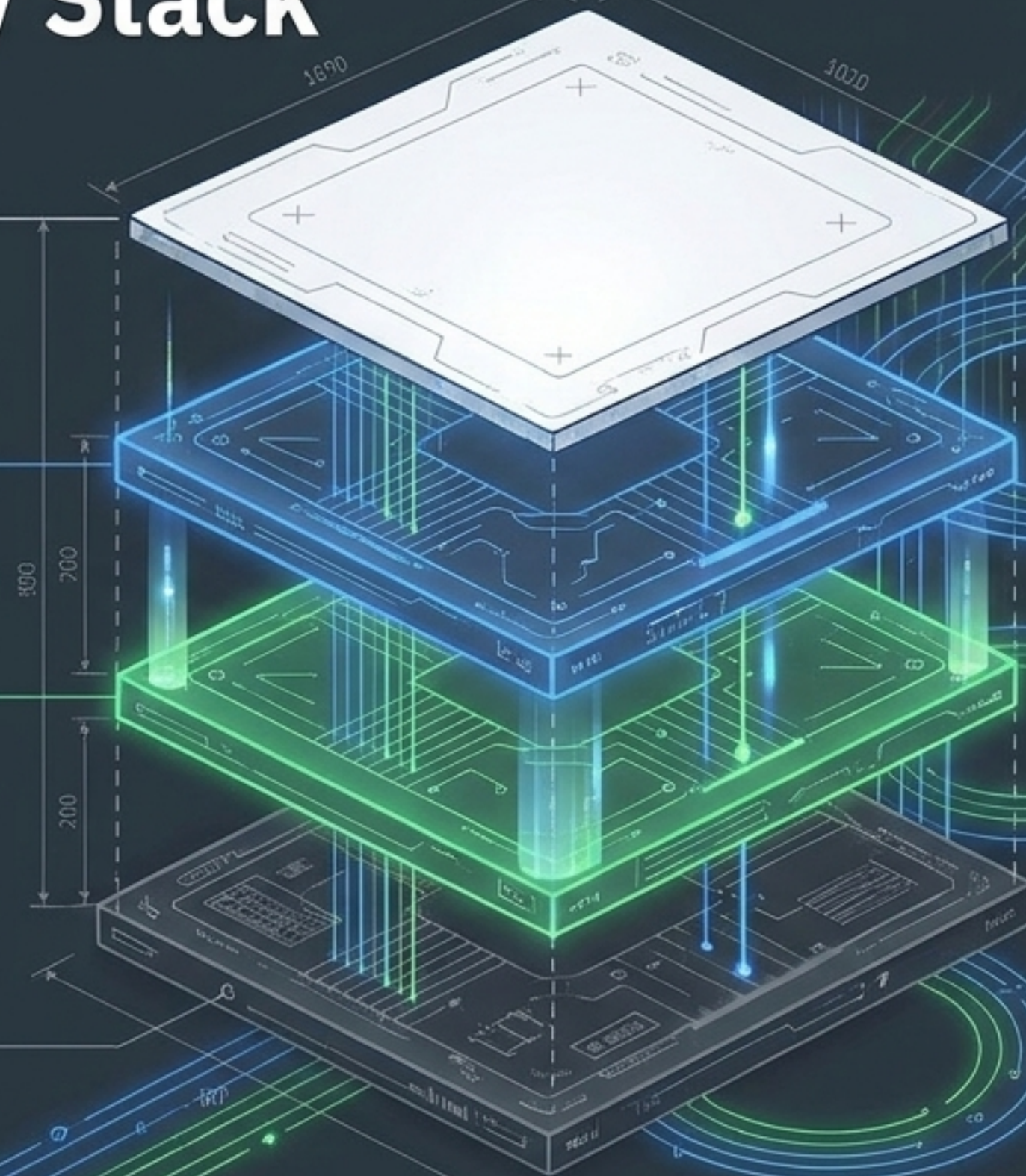
IBM Bob Workbench managing agentic workflows.

Intelligence Layer

Hybrid RAG Architecture.
A keyword/rule engine serves as the rapid primary path, with an optional LLM upgrade path utilising OpenAI's gpt-4o-mini or Granite for complex natural language parsing.

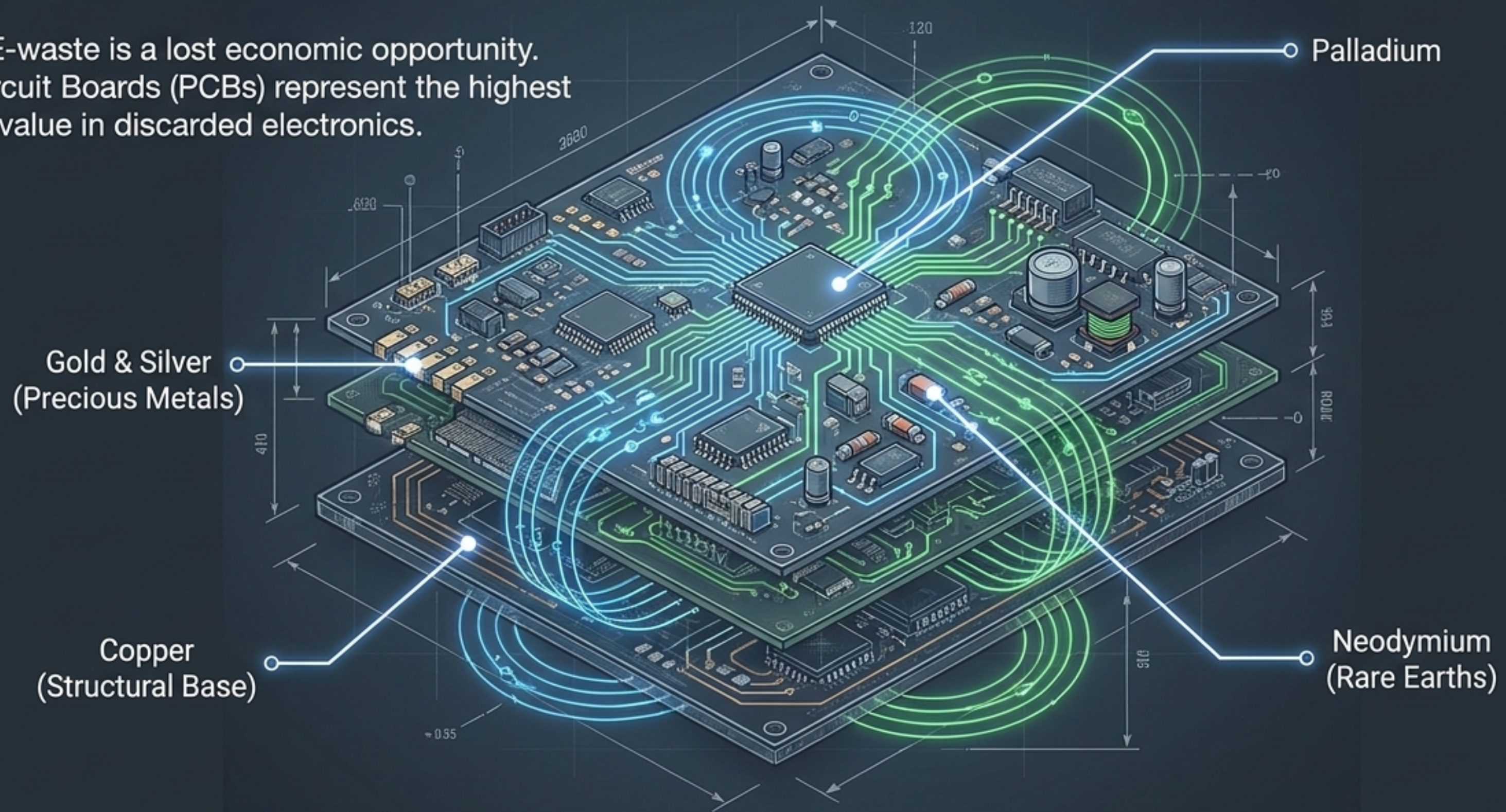
Data Layer

Local JSON repair guides, CSV hub databases, and Pandas for rapid data manipulation.



The Hidden Value in the Circuit Board

Context: E-waste is a lost economic opportunity. Printed Circuit Boards (PCBs) represent the highest economic value in discarded electronics.



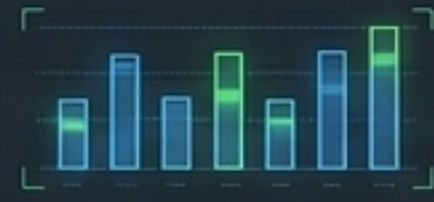
Synthesis Insight: EcoFix AI helps keep these high-value, highly critical materials circulating within the economy, reducing the intense ecological pressure of virgin mineral mining.

Measuring Global Impact (SDG 12)

Direct contribution to Sustainable Development Goal 12: Responsible Consumption and Production.

CO₂ Emissions Saved (lbs)

e.g., averting up to 124 kg of CO₂ by repairing a single television.



Toxic Waste Diverted (lbs)



Landfill Items Avoided



Water Saved (gallons)



Synthesis Insight: Every single device repaired through the platform is a quantifiable, evidence-based victory for the planet.

Closing the Loop

By combining the expertise of the best repair technicians with the highly scalable architecture of Agentic AI, we are making the circular economy a practical, accessible reality for everyone.



EcoFix AI does not just diagnose broken appliances; it repairs the broken relationship between consumers and their electronics.

Repair is the most radical act of sustainability.